

CHE 311 Homework Week 11

1. Flow through a 45° pipe bend. Please calculate the force exerted on the pipe by the fluid. The pressure at the entrance and the exit are $P_1=21$ psi and $P_2=19$ psi, respectively. The volumetric flow rate $Q=3$ ft³/s. The density and viscosity of the fluid is similar as water at room temperature. The length of the pipe is 1 ft. The radius of the entrance and the exit is same, which is 4 in.

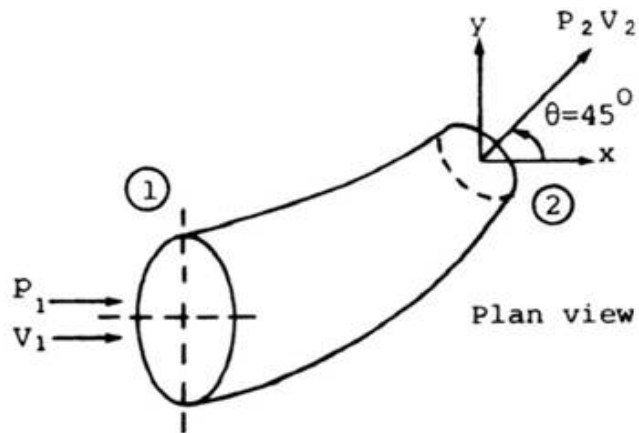


Figure 3. Flow through a 45° pipe bend.

2. BSL 7A.5